QUANTUM FIELD THEORY

These notes are an unoriginal collection drawn from the books and lectures I have studied on this wonderful subject!

Swastik Majumder

Department of Physics

The Ohio State University

September 1, 2026

Contents

Conventions at a glance

Throughout, Greek indices $\mu, \nu, \ldots$ run over spacetime components $0, 1, 2, 3$. THE Latin indices $i, j, \ldots$ run over spatial components $1, 2, 3$, and repeated upper–lower index pairs are summed. We use

$$\hbar = c = 1, \qquad \eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1), \qquad x^\mu = (t, \boldsymbol{x}).$$

The metric choice is sometimes called the *mostly-minus* convention.

1 Classical field theory

This chapter introduces the notation and basic principles used throughout quantum field theory. We begin with natural units, dimensional analysis, and the Lorentz transformation properties of spacetime quantities. The limitations of relativistic single-particle quantum mechanics are then discussed, motivating the transition to quantum fields. Finally, we review classical field theory, the action principle, and the derivation of field equations of motion.

1.1 Units, notation, and dimensional analysis

1.1.1 Natural units

Relativistic quantum theory contains two universal conversion constants: c, and $\hbar$. Setting

$$c = \hbar = 1, \tag{1.1.1}$$

It means that we measure time, length, mass, momentum, and energy using a single base unit and suppress the conversion factors. Thus

$$[E] = [p] = [m], \qquad [t] = [x] = [E]^{-1}. \tag{1.1.2}$$

For restoring ordinary units, two useful identities are

$$\hbar c \simeq 0.19732698 \text{ GeV fm}, \qquad \hbar \simeq 6.58212 \times 10^{-25} \text{ GeV s}. \tag{1.1.3}$$

Consequently,

$$1 \text{ GeV}^{-1} \simeq 0.1973 \text{ fm} \simeq 6.582 \times 10^{-25} \text{ s}. \tag{1.1.4}$$

1.1.2 Mass dimension

It is convenient to write $[X] = n$ when the quantity X has mass dimension n, meaning $X \sim (\text{mass})^n$. Since the phase e^{iS} must be dimensionless,

$$[S] = 0. \tag{1.1.5}$$

In d spacetime dimensions,

$$[\mathrm{d}^d x] = -d, \qquad [\partial_\mu] = 1, \qquad [\mathcal{L}] = d, \tag{1.1.6}$$

because $S = \int \mathrm{d}^d x\, \mathcal{L}$.

The kinetic terms then determine the canonical dimensions of fields. For a real scalar, a Dirac spinor, and a gauge field (all of which will be introduced in time),

$$\mathcal{L}_{\text{scalar}} \supset \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi), \qquad [\phi] = \frac{d-2}{2}, \tag{1.1.7}$$

$$\mathcal{L}_{\text{Dirac}} \supset \bar{\psi}\, i\gamma^\mu \partial_\mu \psi, \qquad [\psi] = \frac{d-1}{2}, \tag{1.1.8}$$

$$\mathcal{L}_{\text{gauge}} \supset -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad [A_\mu] = \frac{d-2}{2}. \tag{1.1.9}$$

Here $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. In four dimensions this becomes

$$[\phi] = 1, \qquad [\psi] = \frac{3}{2}, \qquad [A_\mu] = 1. \tag{1.1.10}$$

If an interaction is written as $g_{\mathcal{O}}\mathcal{O}$, where $[\mathcal{O}] = \Delta_{\mathcal{O}}$, then

$$[g_{\mathcal{O}}] = d - \Delta_{\mathcal{O}}. \tag{1.1.11}$$

This elementary relation anticipates the classification of interactions in effective field theory:

Interaction in $d = 4$	Operator dimension	Coupling dimension
$\frac{\lambda}{4!}\phi^4$	4	$[\lambda] = 0$
$y\,\phi\bar{\psi}\psi$	4	$[y] = 0$
$\frac{g_6}{6!}\phi^6$	6	$[g_6] = -2$
$\frac{c}{\Lambda}\phi F_{\mu\nu}F^{\mu\nu}$	5	$[c/\Lambda] = -1$

Couplings of positive, zero, and negative mass dimension are called relevant, marginal, and irrelevant at the classical level, respectively. Quantum corrections refine this classification; see, for example, Refs. [2, 3].

Example: the dimension of a scalar coupling in general d

Consider $\mathcal{L}_{\text{int}} = -g_n\phi^n/n!$. Since $[\phi] = (d-2)/2$,

$$[g_n] = d - \frac{n(d-2)}{2}. \tag{1.1.12}$$

For $n = 4$, $[g_4] = 4 - d$. Thus ϕ^4 is classically marginal in four dimensions, relevant below four dimensions, and irrelevant above four dimensions.

1.2 Lorentz transformations and index notation

1.2.1 Events, intervals, and the metric

An event in Minkowski spacetime is represented by the contravariant vector

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z). \tag{1.2.1}$$

The Minkowski metric and its inverse are

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{1.2.2}$$

Indices are lowered and raised with the metric:

$$x_\mu = \eta_{\mu\nu}x^\nu = (t, -x, -y, -z), \qquad x^\mu = \eta^{\mu\nu}x_\nu. \tag{1.2.3}$$

Thus an upstairs spatial component and its downstairs counterpart differ by a minus sign. The invariant interval is

$$x^2 \equiv x_\mu x^\mu = \eta_{\mu\nu}x^\mu x^\nu = t^2 - \boldsymbol{x}^2. \tag{1.2.4}$$

Intervals are timelike, null, or spacelike according as x^2 is positive, zero, or negative respectively.

1.2.2 Contravariant and covariant objects

A Lorentz transformation is a linear map

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} \tag{1.2.5}$$

that preserves the metric:

$$\Lambda^{\mathsf{T}} \eta \Lambda = \eta, \qquad \text{or in components} \qquad \eta_{\rho\sigma} \Lambda^{\rho}{}_{\mu} \Lambda^{\sigma}{}_{\nu} = \eta_{\mu\nu}. \tag{1.2.6}$$

An object transforming as in equation (1.2.5) is a contravariant vector. A covector transforms with the inverse matrix:

$$a'_{\mu} = (\Lambda^{-1})^{\nu}{}_{\mu} a_{\nu} = \Lambda_{\mu}{}^{\nu} a_{\nu}. \tag{1.2.7}$$

More generally, every upper index receives one factor of Λ and every lower index one factor of Λ^{-1}.

For a boost of speed v along the x-direction,

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}. \tag{1.2.8}$$

Direct multiplication verifies equation (1.2.6).

1.2.3 Derivatives and index placement

The spacetime derivative is naturally a covector. By the chain rule,

$$\partial'_{\mu} \equiv \frac{\partial}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial}{\partial x^{\nu}} = (\Lambda^{-1})^{\nu}{}_{\mu} \partial_{\nu}. \tag{1.2.9}$$

With the mostly-minus metric,

$$\partial_{\mu} = (\partial_t, \boldsymbol{\nabla}), \qquad \partial^{\mu} = (\partial_t, -\boldsymbol{\nabla}). \tag{1.2.10}$$

The d'Alembertian is the Lorentz-invariant differential operator

$$\Box \equiv \partial_{\mu} \partial^{\mu} = \partial_t^2 - \nabla^2. \tag{1.2.11}$$

1.2.4 Why the dot product is invariant

Let a^{μ} and b^{μ} be Lorentz vectors. Their transformed inner product is

$$\begin{aligned} a' \cdot b' &= \eta_{\mu\nu} a'^{\mu} b'^{\nu} && (1.2.12) \\ &= \eta_{\mu\nu} \Lambda^{\mu}{}_{\rho} \Lambda^{\nu}{}_{\sigma} a^{\rho} b^{\sigma} && (1.2.13) \\ &= \eta_{\rho\sigma} a^{\rho} b^{\sigma} = a \cdot b, && (1.2.14) \end{aligned}$$

where the Lorentz condition equation (1.2.6) was used on the third line. In matrix notation, this is simply

$$a'^{\mathsf{T}}\eta b' = a^{\mathsf{T}}\Lambda^{\mathsf{T}}\eta\Lambda b = a^{\mathsf{T}}\eta b. \tag{1.2.15}$$

Important examples of Lorentz scalars and invariant expressions include

$$x^2 = t^2 - \boldsymbol{x}^2, \qquad p^2 = E^2 - \boldsymbol{p}^2 = m^2, \tag{1.2.16}$$

$$p \cdot x = Et - \boldsymbol{p} \cdot \boldsymbol{x}, \qquad \partial_\mu \phi\, \partial^\mu \phi = (\partial_t \phi)^2 - (\boldsymbol{\nabla}\phi)^2, \tag{1.2.17}$$

$$F_{\mu\nu}F^{\mu\nu} = 2(\boldsymbol{B}^2 - \boldsymbol{E}^2), \qquad \bar{\psi}\psi = \text{Lorentz scalar}. \tag{1.2.18}$$

For proper Lorentz transformations, $\det\Lambda = +1$, the integration measure d^4x is also invariant. Under the full Lorentz group one may more precisely write $\mathrm{d}^4x' = |\det\Lambda|\,\mathrm{d}^4x = \mathrm{d}^4x$.

Not every object with no free index is an ordinary scalar. For example, $\epsilon_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$ is a *pseudoscalar*: it changes sign under improper Lorentz transformations such as parity. This distinction becomes important when discrete symmetries are discussed.

1.3 Why relativistic one-particle quantum mechanics is not enough

In nonrelativistic quantum mechanics, the Hamiltonian of a free particle is $H = \boldsymbol{p}^2/(2m)$. A tempting relativistic replacement is

$$H = \sqrt{\boldsymbol{p}^2 + m^2}, \tag{1.3.1}$$

keeping only the positive-energy branch. This apparently reasonable theory contains a deep localization problem. The discussion below follows the standard motivation given in Refs. [1, 2, 3].

Suppose a particle is perfectly localized at the origin at $t = 0$. In the position basis, the amplitude to detect it at $\boldsymbol{x}$ after time t is formally

$$K(t,\boldsymbol{x}) = \left\langle \boldsymbol{x} | \mathrm{e}^{-\mathrm{i}Ht} | \boldsymbol{0} \right\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \exp\left[\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x} - \mathrm{i}\sqrt{\boldsymbol{p}^2 + m^2}\, t\right]. \tag{1.3.2}$$

If a strictly localized relativistic particle respected naive one-particle causality, this amplitude would vanish whenever $r \equiv |\boldsymbol{x}| > t$. It does not.

To see the origin of this result cleanly, introduce the positive-frequency two-point function

$$\Delta^+(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{2E_{\boldsymbol{p}}} \mathrm{e}^{-\mathrm{i}E_{\boldsymbol{p}}t + \mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}, \qquad E_{\boldsymbol{p}} = \sqrt{\boldsymbol{p}^2 + m^2}. \tag{1.3.3}$$

The kernel in equation (1.3.2) satisfies

$$K(t,\boldsymbol{x}) = 2\mathrm{i}\,\partial_t \Delta^+(x). \tag{1.3.4}$$

For a spacelike separation, define

$$\rho \equiv \sqrt{r^2 - t^2} > 0. \tag{1.3.5}$$

The momentum integral can be evaluated in terms of a modified Bessel function:

$$\Delta^+(x) = \frac{m}{4\pi^2\rho} K_1(m\rho), \qquad x^2 < 0. \tag{1.3.6}$$

Since $K_1(z)$ is nonzero for positive z, both $\Delta^+(x)$ and, for generic t, the transition kernel have an exponentially small but nonzero spacelike tail. At $m\rho \gg 1$,

$$\Delta^+(x) \sim \frac{\sqrt{m}}{4\sqrt{2}\,\pi^{3/2}\rho^{3/2}} e^{-m\rho}. \tag{1.3.7}$$

The characteristic distance is the Compton wavelength m^{-1}. Attempting to localize a particle on shorter scales requires momenta large enough to make particle production energetically possible, so a fixed-particle-number description has left its domain of validity.

1.4 Classical fields and the action principle

1.4.1 From coordinates to fields

In point-particle mechanics, the dynamical variables are a finite collection of generalized coordinates $q_a(t)$. In field theory, the dynamical variables are fields $\phi_a(x)$, one value at every spacetime point. The discrete label a may distinguish species or internal components; it is not a spacetime index unless explicitly stated.

For a local first-derivative theory, the Lagrangian density is

$$\mathcal{L} = \mathcal{L}(\phi_a, \partial_\mu\phi_a; x), \tag{1.4.1}$$

and the action is

$$S[\phi] = \int_\Omega d^d x\, \mathcal{L}. \tag{1.4.2}$$

The Lagrangian density may change by a total derivative without changing the bulk equations of motion:

$$\mathcal{L} \longrightarrow \mathcal{L} + \partial_\mu K^\mu. \tag{1.4.3}$$

The action then changes only by a boundary term. Boundary terms are not always physically irrelevant—they matter for boundaries, conserved charges, and gravity—but they do not modify the local Euler–Lagrange equations when the variations obey the assumed boundary conditions.

1.4.2 Deriving the field Euler–Lagrange equations

Consider an arbitrary infinitesimal variation

$$\phi_a(x) \longrightarrow \phi_a(x) + \epsilon\,\delta\phi_a(x), \tag{1.4.4}$$

with $\delta\phi_a = 0$ on the boundary $\partial\Omega$. To first order,

$$\delta S = \int_\Omega d^d x \left[\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\partial_\mu(\delta\phi_a) \right] \tag{1.4.5}$$

$$= \int_\Omega d^d x \left[\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right) \right]\delta\phi_a + \int_{\partial\Omega} d\Sigma_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\delta\phi_a. \tag{1.4.6}$$

The boundary term vanishes by assumption. Since the remaining $\delta\phi_a(x)$ are arbitrary, stationarity $\delta S = 0$ implies

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right] = 0. \tag{1.4.7}$$

This is the field-theory Euler–Lagrange equation. It is the direct analogue of its point-particle counterpart,

$$\frac{\partial L}{\partial q_a} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}_a} \right) = 0. \tag{1.4.8}$$

1.4.3 What if the Lagrangian contains higher derivatives?

There is no mathematical rule forbidding a Lagrangian density from depending on higher derivatives. If

$$\mathcal{L} = \mathcal{L}(\phi_a, \partial_\mu \phi_a, \partial_\mu \partial_\nu \phi_a), \tag{1.4.9}$$

two integrations by parts give

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} + \partial_\mu \partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \partial_\nu \phi_a)} = 0. \tag{1.4.10}$$

The appropriate boundary conditions now fix both $\delta\phi_a$ and its first normal derivative. For derivatives through order N, the compact result is

$$\sum_{n=0}^{N} (-1)^n \partial_{\mu_1} \cdots \partial_{\mu_n} \left[\frac{\partial \mathcal{L}}{\partial(\partial_{\mu_1} \cdots \partial_{\mu_n} \phi_a)} \right] = 0, \tag{1.4.11}$$

where the $n = 0$ term means $\partial \mathcal{L} / \partial \phi_a$.

Why, then, do introductory theories usually contain at most first derivatives in $\mathcal{L}$?

- A nondegenerate Lagrangian with higher time derivatives generally yields higher-order equations of motion, requires extra initial data, and suffers from the Ostrogradsky instability: its Hamiltonian is unbounded below.
- Additional derivatives increase operator dimension, so such terms are often suppressed by powers of a high scale in an effective field theory.
- First-derivative actions already generate second-order equations, which supply the familiar relativistic wave propagation problem.

These statements have important qualifications. Total derivatives do not add dynamics; constrained or degenerate higher-derivative theories can evade the Ostrogradsky conclusion; and higher-derivative operators are routinely used perturbatively in effective field theory.

1.5 Examples of Euler-Lagrange equation of motion

1.5.1 A real scalar field

Consider

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi). \tag{1.5.1}$$

The two Euler–Lagrange derivatives are

$$\frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi), \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi. \tag{1.5.2}$$

Therefore,

$$\Box \phi + V'(\phi) = 0. \tag{1.5.3}$$

For $V(\phi) = m^2\phi^2/2$, this reduces to the Klein–Gordon equation

$$(\Box + m^2)\phi = 0. \tag{1.5.4}$$

A plane wave $\phi \sim \mathrm{e}^{-\mathrm{i}p\cdot x}$ satisfies this equation only if

$$p^2 = m^2, \qquad \text{i.e.} \qquad E^2 = \boldsymbol{p}^2 + m^2. \tag{1.5.5}$$

Thus Lorentz invariance of the action directly produces the relativistic dispersion relation.

For the interacting potential

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4, \tag{1.5.6}$$

the equation becomes

$$(\Box + m^2)\phi + \frac{\lambda}{3!}\phi^3 = 0. \tag{1.5.7}$$

1.5.2 A complex scalar field

Let

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - \frac{\lambda}{2}(\phi^*\phi)^2. \tag{1.5.8}$$

Although ϕ^* is the complex conjugate of ϕ, they are treated as independent variables during variation (We will later see that a complex field is actually two real scalar fields). Varying with respect to ϕ^*,

$$\frac{\partial \mathcal{L}}{\partial \phi^*} = -m^2\phi - \lambda(\phi^*\phi)\phi, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = \partial^\mu \phi, \tag{1.5.9}$$

and hence

$$(\Box + m^2)\phi + \lambda(\phi^*\phi)\phi = 0. \tag{1.5.10}$$

Varying with respect to ϕ gives the complex-conjugate equation. The Lagrangian is invariant under the global transformation $\phi \to \mathrm{e}^{\mathrm{i}\alpha}\phi$, which will lead to a local and a global conservation law as given by Noether's theorem.

1.5.3 The electromagnetic field

The gauge potential A_μ defines the antisymmetric field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{1.5.11}$$

In the presence of an external current j^μ, take

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - j^\mu A_\mu. \tag{1.5.12}$$

The Lagrangian depends on A_ν through the source term and on its derivative through $F_{\mu\nu}$:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = -j^\nu, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -F^{\mu\nu}. \tag{1.5.13}$$

The Euler–Lagrange equation is therefore

$$\partial_\mu F^{\mu\nu} = j^\nu, \tag{1.5.14}$$

which contains Gauss's law and the Ampère–Maxwell law. The remaining two Maxwell equations follow identically from the definition of $F_{\mu\nu}$:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0. \tag{1.5.15}$$

Taking ∂_ν of equation (1.5.14) gives

$$\partial_\nu j^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0, \tag{1.5.16}$$

because the derivatives are symmetric in μ, ν while $F^{\mu\nu}$ is antisymmetric. Current conservation is thus required for consistency.

The action is unchanged under

$$A_\mu \longrightarrow A_\mu + \partial_\mu \alpha, \tag{1.5.17}$$

provided the source is conserved (up to a boundary term). This gauge redundancy will later be central to quantizing the electromagnetic field.

References

[1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Addison–Wesley, Reading, Massachusetts, 1995).

[2] M. Srednicki, *Quantum Field Theory* (Cambridge University Press, Cambridge, 2007).

[3] M. D. Schwartz, *Quantum Field Theory and the Standard Model* (Cambridge University Press, Cambridge, 2014).

[4] D. Tong, *Lectures on Quantum Field Theory*, University of Cambridge Part III Mathematical Tripos lecture notes, `https://www.damtp.cam.ac.uk/user/tong/qft.html`.

[5] T. J. Osborne, *Lectures on Quantum Field Theory*, Leibniz University Hannover. Lecture notes transcribed by A. V. St. John, `https://github.com/avstjohn/qft`.

[6] P. Ramond, *Field Theory: A Modern Primer*, 2nd ed. (Addison–Wesley, Redwood City, California, 1989).

[7] T.-P. Cheng and L.-F. Li, *Gauge Theory of Elementary Particle Physics* (Clarendon Press, Oxford, 1984).

[8] S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations* (Cambridge University Press, Cambridge, 1995).